

Caring about Content about Care: Building Quality Content with Clinicians

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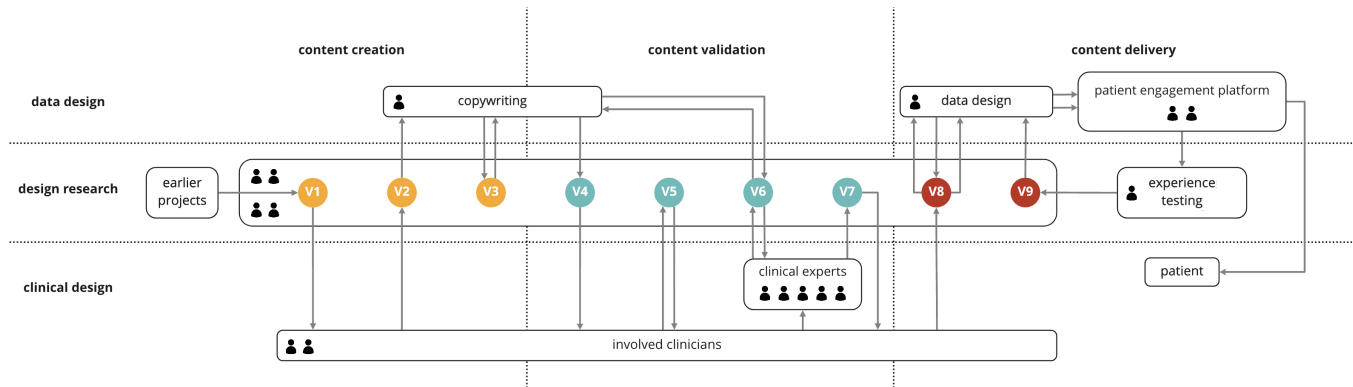


Figure 1: An outline of the content design process, moving from the content creation phase into the content validation phase, and finally into the content delivery phase

ABSTRACT

Medical care outside the doctor's office and the hospital is gaining traction. Lifestyle programs for transmural and remote care are everywhere, facilitated by hospitals as part of rehabilitation, by general practitioners as preventative measures, and more and more commonly, by various private (health) organizations through app stores. What might often be overlooked is the time and energy that is spent on the creation of the content in these programs. In this article, we discuss the elaborate content creation process of lifestyle content for an outpatient clinic for patients scheduled for an ablation surgery. We describe the close collaboration between the clinicians and our design research team, and reflect on how to streamline and formalize this process.

CCS CONCEPTS

• **Human-centered computing** → **User studies; Field studies.**

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1 INTRODUCTION

Healthcare information consumption is changing. With the many platforms that are currently available for engagement with one's own health, different types of organizations are competing for patients' attention: from commercial players in the quantified self area and social networks to healthcare providers, insurances and government agencies. Especially traditional healthcare organizations are reinventing themselves to reach their patients in increasingly creative ways, while private companies offer mainly supplemental solutions such as meditation app Headspace [7] and Apple Health [6]. Simultaneously, the channels where healthcare professionals share health information are changing. Online patient platforms are becoming more common, and recent developments in content management in clinical settings have seen clinicians move to platforms like Instagram and TikTok to reach wider audiences [2, 14].

Healthcare platforms are characterized by high complexity, regulation and entanglement of different organizations and services.

Designing and implementing new healthcare solutions is a multifaceted challenge. In this work, we wish to spotlight a particular facet: content and dealing with content in a multi-organizational design process. In conceptual design projects, system content is frequently neglected and not taken as equally important as other solution aspects such as user interface, customer journey, and functional integration with platforms and services. Often a system infrastructure is created, with some mock-up messages or a single conversation flow, the “happy path”. More often than not, this is where content design and validation stops. In some cases, content can be developed at later stages in the process as long as insights about target group and messaging approach mature. However, when we are prototyping for behaviour change or longitudinal use, it is important to not only test that the idea conceptually works, but also that the content is reaching the goal that it was designed to reach. This deeper validation requires looking at whether the content reaches the right audience, whether it communicates effectively and whether it has variation, yet also consistency, throughout. After all, the content delivery might need a complete redesign if it turns out that the content should be formatted very differently than the current conceptual design proposes. In practice, the content creation process is complex with a workflow of iterations, loops, negotiation and the occasional backtracking. Why? Because in a healthcare setting, content makes or breaks the solution: in conceptualization, various stages of internal and regulatory approval, usage—or litigation.

The process towards meaningful clinical content creation is complex and deserves a more prominent role within design practice [13]. Currently, there are good examples of work that focused on content creation and validation [20] and content delivery [19, 23], but little work has explored the content design process as a whole [10]. In this paper we describe in detail the content creation process for a large clinical trial with patients who are scheduled for an ablation surgery. We elaborate on the different steps of the process towards our content database and detail the challenges we faced along the way. We conclude with insights on how to formalize and better streamline the clinical content creation process.

2 BACKGROUND

Healthcare systems all over the world have been under severe pressure for years, and innovation is necessary in many areas. In a quest for more predictive, precise, preventive and personalized healthcare [5], healthcare designers are looking for ways to effectively use data [1].

Challenges in the healthcare domain are becoming more complex, holistic and multidisciplinary. This can be seen in an increase in the application of co-creation, co-design, participatory design and data-enabled design processes [9, 21] applied to healthcare problems and emerging medical systems. Increasing complexity of multi-stakeholder processes also necessitates that design practice assumes a bigger role in healthcare innovation, and that clinicians working on these processes are developing their design skills in this process as well. These types of clinical design projects also call for new design approaches, with a focus on infrastructure [12].

The value of lifestyle coaching for healthcare has been established broadly in the last years. Lifestyle healthcare programs are increasing in popularity, both in hospitals, but also in primary care

and on the consumer market. Lifestyle coaching is by now widely available on the private market, with different coaches focusing on different topics, ranging from nutrition advice during pregnancy, to weight loss programs and digital detox. All of these programs rely heavily on content: information and notifications to be sent out to motivate individuals to change their behaviour in one way or another. These include app designs [26], persuasive design in public spaces [3, 4], and personal objects to promote a healthy lifestyle [15]. The consumer market has also been buying into this movement, with the wide adoption of health tracking devices such as Fitbit [18], Apple Health [6] and apps like Strava [24].

3 METHOD

Atrial fibrillation (AF) is a cardiac condition that makes the heart beat rapidly and irregularly over varying periods of time. It makes patients anxious, stressed and has a big impact on their quality of life. Two percent of the world population is affected by atrial fibrillation. The purpose of the content that was designed within this project was to provide guidance in lifestyle changes associated with the reduction of AF symptoms. A total of 150 patients will be included in the trial, half of which in the treatment group. At time of writing, this clinical trial is ongoing at Catharina Hospital in Eindhoven, the Netherlands. The included patients are scheduled for an ablation surgery to reduce their AF symptoms. During the clinical trial, they will receive personalized lifestyle content through the online Philips Engage¹ platform. A short demo of the Engage platform can be found in Video Figure 1. The content they receive is based on which topics their nurse practitioner considers the biggest risk factors for their AF symptoms. When we refer to content in this paper, we are primarily speaking of text messages of around 50-200 words with attached images, links and (external) videos. This lifestyle program will delay their ablation surgery for up to six months, but the lifestyle change will likely already alleviate their symptoms as well as increase the chances of success for the surgery.

The targeted solution is a patient engagement platform. Participants will use the platform on a daily basis to gain insight into a healthy lifestyle and to keep track of their personal lifestyle goals (as discussed with their nurse practitioner in consults). The platform supports notifications, setting and logging goals and sending direct messages to healthcare professionals. A crucial aspect of the platform is the presentation of personalized content. The platform was extended with respect to the design of new content sequences specifically developed for AF patients, referred to as lifestyle protocols. These protocols need to be authored and aligned with the content. Based on the protocols the platform triggers this scheduled content according to the stage of treatment that the patient is in. Content and content protocols need to be consistent and this requirement is specifically checked in a quality assurance measure. The selection of risk factor related protocols that contain educational content is personalized for each patient, based on the assessment of the nurse practitioner in the intake consult.

¹<https://www.philips.nl/healthcare/sites/vitalhealth/products/engage-patientportaal-solutions-platform>

All content for the protocols was collected and written by a core team of three designers (one data designer and two design researchers), a copy writer and two clinicians: the nurse practitioner who would primarily run the outpatient clinic and the clinical PhD student who was also in charge of setting up the clinical study procedures. Besides this core team, two other design researchers worked on the content protocols during the process. The designers were responsible for setting up the content structure and writing the first versions of the articles, the clinicians checked the clinical accuracy and alignment with the existing care path and the copywriter made sure that all language used in the articles was appropriate for the diverse audience (B2 level Dutch).

We were fortunate to work with two clinicians who had the time to be actively involved in the content creation process. For almost an entire year, we met each other on weekly basis for an hour. On top of that, they spent many hours reviewing the content that was drafted, provided feedback and reviewed the many revisions. Additionally, we were supported by the Engage team in the configuration of the platform for the needs of the clinical trial. Our team in close collaboration with the platform's consultants uploaded the content onto the platform, streamlined the interaction between the content and questionnaires sent out through the same platform and set up the logic for sending out the protocols.

An overview of the process can be found in Figure 1. The process started with the content creation phase, during which the outline for the content was initially created by the design research team, based on interview data and experience with previous content projects (e.g. [8, 17]). This outline was discussed with the involved clinicians to further fine-tune the amount and the length of the protocols and to set up the outline for each individual content item in a draft version. This first version of the content items was then shared with a qualified copywriter to maintain a consistent tone of voice and to make sure that the messages were also understandable for those patients for whom Dutch is their second language, and those with lower literacy. Consequently, all messages were composed in Dutch at B2 level. The design research team then reviewed the content items from the copywriter, checking whether the essence of the original message was captured correctly. In some cases, feedback was then sent back to the copywriter, which would result in (minor) alterations.

We then entered the validation phase. In this phase, the content items were checked by the clinicians. They checked the medical accuracy of the messages based on their expertise, as well as whether the items were appropriate for the patient population. In some cases, additional clinical experts were involved when the content lay outside of the involved clinicians' expertise (e.g. a pulmonologist and a dietician). In most cases the proposed changes during this phase were minor and could be implemented by the design research team. In other cases the copywriter was consulted again to do a full rewrite of a content item.

After all content items were approved by the clinical team and all members of the design team (copywriter included), we moved into the delivery phase. During this phase, the design researchers and data designer aligned all protocols with regards to timing of the messages and make sure they were ready for implementation into the software platform. This included linking the right images to the right items and making sure that all settings for the content delivery

were set correctly for each protocol (e.g. the date the content would become available, whether it was a task or an information item and when the item expired).

4 FINDINGS

In total, 200 content items were created, spread over 12 different protocols. Five of these protocols are sent to all patients in the treatment group, which we refer to as general protocols. The other seven protocols are activated based on their relevance. We refer to those as the risk factor protocols, as each of them targets a specific risk factor through a protocol that aims to decrease the risk of atrial fibrillation episodes. These risk factors are high blood pressure, unhealthy nutrition, high cholesterol, sleep apnea, poor physical activity, alcohol, and smoking.

In this section we describe challenges we faced throughout the different phases of the process and insights that we gained in relation to these phases, and how this can be generalized for other content-rich design processes. The core team spent a total of 7.5 person months (by approximation) on the creation of this content over the course of a year. This includes setting up the structure for the protocols, collecting information for the protocols, writing the content items (including review and revision) and uploading the content to the Engage platform. It has to be noted that a lot of prior work was available from previous studies [8, 17], so the cumulative time investment is even higher.

4.1 Content creation

In the initial phase, the most important task for the team was to get an insight into the care pathway that AF patients currently experience in the hospital. More than ten sessions were spent on getting the detail of this right, as the timing of the content items would define their relevance to the patient at the time. Especially as we wanted to prepare them for their face-to-face consultations with the content items, we needed to understand when they would be visiting the hospital and what they would be discussing with the medical professionals. While there was a protocol for this, in we quickly learned that in practice there would be quite some variation in the timing of these meetings, based on clinician and patient availability, as well as external factors such as national holidays. On top of that, the design process took place during the COVID-19 pandemic, which meant that the situations in the hospital were prone to change all the time. Through many versions of the care pathway, we were able to understand the exact journey that the patient would take over the course of their AF treatment, with some side notes with regards to the timing.

At the same time, we decided on the tone of voice and the type of content that would be shared with the patients, in collaboration with the copywriter. The tone would be formal, yet friendly and supportive. The messages ought to focus on providing factual information about AF and its risk factors and on giving the patients practical tips on how to improve their lifestyle.

All content within the system was written in B2 level Dutch. The copywriter was experienced with writing content at this level but occasional conflicts did arise. During some expert feedback sessions, the clinicians expressed that they considered some of the content to be pedantic and that it did not fully acknowledge what

the patients would already know. Unfortunately, limited time and limited platform functionality meant that it was difficult to personalize at this level, and we chose instead to focus the personalization on the topics of the content rather than the tone of voice. As the tone of voice had to be generalized for all patients, we had to take into consideration that some of the patients did not take in this information before, or would benefit from repetition. Oversimplification, on the other hand, could also lead to clinical inaccuracy or statements that came across as too bold. Here, too, there was a conflict with regards to engagement, as we knew from previous studies that a more colloquial tone resonated with many patients.

4.2 Content validation

To ensure that the protocols and the individual content items were clinically appropriate for our patient population and to make use of the vast body of knowledge with regards to AF as a condition, the involved clinicians reviewed all protocols and content items multiple times. They participated in weekly project meetings, as well as work sessions for the content. While we initially feared that some of these activities might be too time consuming next to their other tasks in the hospital, we found that they were very willing to free up the time as they saw how the time investment would benefit their patients and contribute to better care standards in the long term.

Due to the COVID-19 pandemic, much of our collaborative work took place online, which proved very fruitful as the clinicians could easily provide their feedback on our work in a set of Miro [16] boards. We worked asynchronously on the content items, which meant that they could fit it into their day-to-day routine when they had the time.

Our clinicians would often pick up the organizational tasks within the hospital when required, and arrange for meetings where part of our design research team, as well as the clinicians themselves, were present. The clinicians on our team could then easily discuss the clinical content of the messages in their medical jargon, while we could pick up remarks regarding the design of the system (e.g. timing of the content, relevant data to collect, etc.). These experts included dietitians, a nurse practitioner of a cooperating hospital, a pulmonologist and a sleep specialist of a cooperating hospital. The experts were asked to review only specific protocols that were relevant to their expertise.

We see the fluent collaboration with the clinicians as a highlight and an important requirement for successful implementation of our data solutions in practice. By collaboratively designing the care pathway, it was much easier for us to anticipate exactly the situation for the patients and the implementation of our envisioned system into the hospital. Having close contact with these two clinicians also made it much easier for us to reach other relevant experts within the hospital in a timely fashion.

An important difference between this process and earlier work around healthcare content was that the role of lead investigator was with the hospital, rather than with us, contrary to what was the case for [17]. This meant that there was less freedom to make major changes to the situation from a design perspective, as the end goals of the study are also clinically oriented. While this meant that we had to take a step back with the amount of technological

innovation that we could implement, it also resulted in a smoother process with the medical ethical committee.

4.3 Content delivery

After all content was finished and validated by the clinicians, our team picked it up to turn it into lifestyle protocols that can be delivered to the patients via the patient engagement platform, Philips Engage. First of all, all content needed to be transferred from our team's work environment to an understandable format for the platform. Unfortunately the platform does not support automatic input via an API therefore our data designer set up a script that would execute the required manual tasks at a higher speed and with more precision than we could. After this step was done the data designer configured all the protocols in a way that our design team could carry out an experience testing activity. One of our design researchers acted as a participant. This way we could test whether the content was adequately scheduled and could once more read the content from the patient perspective. When required the data designer updated the content on the platform based on this testing. Due to the novelty of this system for the present clinical context, this unfortunately coincided with various error messages and the occasional software bug. In an article on Data-Enabled Design in the clinical context similar obstacles were highlighted [17].

Following the recommendations laid out in the same article [17], the team identified which elements are safely automatable and which elements require manual tasks from the participating healthcare professionals or the design researchers. Based on this analysis we concluded that delivering the pre-scheduled content for the participating patients can happen automatically by the Engage platform.

In a collaborative effort, the (design) researchers and the healthcare professionals identified all the relevant dates and times when lifestyle coaching content could be presented at the right moment. These moments are fully based on the timeline of the redesigned care pathway. The participating patients are notified by Engage when there is a new content item available that they can read via the platform. The content is most optimally presented via a web browser in a task list. The participant opens the content item, and can spend as much time as they require to read the text, can click on the embedded links, and watch included videos. Once they are done with exploring the content they click on the button that they read the content item. This action is logged by the platform. The healthcare professionals are also able to track via the Engage platform whether the patients read or ignored the scheduled content items. The redesigned care pathway includes several face-to-face consultations with a healthcare professional. These are suitable moments to discuss the progress of the patient, and the relevance and feasibility of the content items the patient received. Moreover, at the end of each lifestyle protocol the design research team added an evaluative questionnaire for the patients about the content they received. This way our approach is following the requirement to have evaluation data present when designing health communication [11].

When it comes to content delivery, we realised that unfortunately at this time we were limited to presenting the content items through an online platform. Ideally, this would be more strongly integrated into the everyday life of the patients and be available on

the platform they prefer to use. Additionally, data tracking could be more strongly integrated as in [8], so that the content can be delivered based on data triggers.

5 DISCUSSION

Overall, the collaboration with the team on the content for the protocols was very rewarding. The spirit in the team was good, values felt aligned, and each of us had the same end goal in mind. The multidisciplinary character of the team was vital to this successful collaboration. In this specific case, the expertise of all people involved was required. In order to achieve personalized healthcare down to the level of the content that is provided, this detailed level of content creation was required. However, reflecting on the process, we do not expect this to be feasible in every clinical specialization for each care pathway. Therefore, we want to take the opportunity to reflect on the separate elements of the process to see where they might be streamlined and automated to create a more fluent flow from a protocol outline to the content on the patient's device.

First, the flow of the process could still be greatly improved. As we did not fully grasp the scope of the challenge ahead of us from the beginning, we would often have to divert from the logical path that we had set out for ourselves as some protocols needed a different approach than others. Content items then often went back and forth between different members of the content team, which meant that there would be synchronization issues between different versions. The online collaboration tools we were using did not always prove optimal for managing these versions.

Secondly, one of the biggest hurdles in the process was that the Engage platform we were using to present the content to the patients was not optimized for the way in which we intended to use it. Previous use of the platform had seen less extensive content protocols, and no interlinked protocols. Although extensive research has been done into this type of software, there were many solutions that we could not use due to a lack of clinical approval, data security concerns and technical issues with integrating them into everyday care with high participant numbers. This meant that we had to come up with some wizard-of-oz solutions to carefully time the content items to the face-to-face visits that the patients would have with their nurse practitioner. While for patients this still means they will have a streamlined experience, it introduced manual tasks for the nurse practitioner in enabling and disabling protocols. Unfortunately the clinical staff experience will therefore not entirely simulate the workload reduction for the clinical staff.

Finally, we ran into the issue that the amount of personalization that could be achieved with the available resources was still limited. While we would have liked to have patient-specific content that was tailored to their personal situation like in [8], this was difficult to scale up within the current clinical context. We ran into the situation where creating more personalized content meant creating more versions of the content. While we did this for some situations (i.e. two different protocols for either an internal program or a more extensive program at a specialized facility), this was limited to situations where the information would otherwise be factually incorrect (e.g. stating the wrong location or specialist for procedures). In the current setup, each step of personalization would exponentially increase the number of content items that needed to be prepared.

5.1 Future opportunities

The future of care that we envision holds many of the personalized care programs like the ones we used in this project. Therefore, we consider it very important to create a dedicated content creation workflow that helps achieve this in many places, with many different teams. In our case, the clinicians proved to be the experts, and while many of the other tasks in the process might be replaced, we believe that their role cannot and should not be replaced by other stakeholders or even technology. The first challenge to tackle would be the manual labour associated with uploading the content to the content platform. This process could be easily automated by making the platform such that it can automatically import an Excel sheet, without requiring manual interference from the design team. An online platform could also help to keep track of changes to content items and protocols. Sharing the content with other team members online through Miro [16] worked for this project, but we encountered issues with different versions being spread to different members of the team and not fully converging in the end.

5.1.1 Tools to support content creation. Content creation is an area of opportunity for the HCI community to research and for specialized tools be built and tailored. While the content might not always be the core contribution of healthcare platform design, it is extremely important that the process of content creation, validation and delivery is smooth—in all its complexity. This requires a redesign of the platform which is optimised for quickly editing both individual items and the protocols as a whole. Additionally, having a review feature within this platform, rather than a separate platform entirely, would also significantly streamline the validation process.

5.1.2 Dynamic content. A huge issue that we encountered was the sheer amount of content items in order to reach the level of personal attention and understanding that the clinicians intuitively offer in face-to-face consults. We wanted our system to be an extension of that type of interaction, but it was impossible to reach that nuance without an exponential growth of the content database. If content items could be created dynamically, where variables are saved that influence the content of the messages (much like the interactive storytelling editor Inky does [25]), far less content items would need to be created and sent, which would also lower the burden of treatment for patients as they receive less notifications.

5.1.3 Personalization of content delivery. Besides personalization through offering different protocols and content items, we also see great potential in personalization of the type of content (e.g. the medium, the length of the text and the tone of voice). In our experience, it was difficult to write content in a way that would be engaging for *any* kind of audience, and we often went back and forth between a more educational tone that assumed little prior knowledge, and a more academic tone that presented facts based on medical research. As there is a large difference in how people prefer to receive information, based on e.g. their personality [22], we believe that adding this level of personalization could also increase patient engagement and treatment effectiveness.

5.1.4 Preventative care. An open issue in providing preventative care is how to deliver it to the right people at the right moment.

Especially primary prevention can be difficult as they have usually not had any contact about their health with their doctor yet. It is important to strategically brand the lifestyle content such that it reaches and appeals to a wider audience than just to diagnosed patients. Health insurance could play a role in this, or it could be implemented in consumer products rather than patient platforms. However, it is extremely important to bear in mind that the clinical connection is maintained as it is currently difficult for clinicians to rely on consumer products that have not been certified for clinical use and offer comprehensive, secure and integrative data sharing features.

6 CONCLUSION

In this article we report on a content design and validation process in the context of studying a novel healthcare solution for prospective surgery patients. In outlining the process and providing in-depth experience reports on various aspects of creation, validation and delivery, we have highlighted some challenges in our content design process. We have identified areas for future improvement of the content design process, with possibilities of using technology to streamline the content creation and validation process, to move towards more dynamic content items, to personalize not just content items but also content delivery and to approach lifestyle content for a bigger audience spanning beyond the hospital walls.

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